Calibration-Free P300 Decoding: Cross-Subject Transfer on BrainInvaders2013a

Birzhan Zhunusbekov, Arman Abakov, Nazerke Khamzina, Bekmurat Amangeldiyev
ROBT613 Midterm Project

Abstract—We test whether a linear P300 decoder trained on other users can rank target flashes above chance and how much it loses relative to subject-specific calibration. The study uses 17 single-session BrainInvaders2013a subjects, two training runs per subject, and a fixed five-channel physiological montage. A common pipeline extracts 50 ms ERP window means, scales features on training data, and fits shrinkage linear discriminant analysis. Within-subject, leave-one-subject-out (LOSO), and class-count-matched LOSO conditions share the same held-out test epochs. Mean subject-level ROC AUC is 0.875 ± 0.056 for Within, 0.724 ± 0.071 for LOSO-matched, and 0.810 ± 0.072 for LOSO. Directional subject-level comparisons use Holm-adjusted Wilcoxon tests. The study is retrospective and pilot-informed. It establishes offline flash-level performance in this cohort, with no claim of online operation or exact permutation inference for temporally dependent epochs.

Index Terms—P300, EEG, cross-subject transfer, shrinkage LDA, grouped evaluation

I. Question and Practical Significance

A calibration phase asks a new BCI user to supply labelled examples before interaction. Reducing that burden motivates generic decoders, but replacing a user's own data with other people's recordings changes signal distributions. We ask whether a simple, interpretable decoder still transfers when its preprocessing and feature representation stay fixed. The question concerns the cost of transfer rather than the highest attainable classification score.

Prior work motivates this question without supplying our numerical result. Barachant and Congedo discuss generic initialization and continuing adaptation in a P300 system using information geometry [2]. Kim and Kim report a zero-training system with a pretrained xDAWN filter and convolutional network [3]. Those methods provide context only. Our core implementation uses the ERP and linear-classifier tools in workshop notebooks 00–06 [4]. We do not implement the later spatial or nonlinear methods, and their device-control accuracy is not directly comparable to our single-flash ROC AUC.

The original hypothesis has three parts. H1 states that LOSO AUC exceeds 0.5 on unseen subjects. H2 states that Within exceeds LOSO on the same test runs. H3 states that Within still exceeds LOSO when both training sets have the same class-specific epoch counts. H3 addresses training-set size as an explanation for a transfer gap. It does not isolate all physiological and recording differences between participants.

A. Retrospective Development Boundary

Subjects 8–10 informed the earlier development run. Before executing the full cohort, we fixed a physiological montage of Cz, P3, Pz, P4 and Oz. The earlier implementation pooled bad-channel flags over all subjects, allowing a held-out subject to affect the training feature set. The revised implementation removes that dependency entirely. It records the pooled flags for audit only. All three conditions use the same five sites and 16 time windows, giving 80 features. This is a disclosed protocol revision, not a claim of prospective preregistration. A sensitivity summary excludes the three development subjects from evaluation.

B. Study Logic

Each of the 17 participants contributes two runs. One run tests and the other trains the Within model. A LOSO model trains on both runs of all other participants and tests on each of the held-out user's runs. A matched model uses a seeded subset of the same external pool, with target and non-target counts equal to Within. Averaging ten matched draws reduces dependence on a single random subset. The test epochs and outcome metric remain identical across these conditions.

II. Data Audit and Signal Inspection

The acquisition report describes a visual P300 experiment recorded in 2013 with 16 scalp electrodes, a g.USBamp amplifier and a left-earlobe reference [1]. We retain subjects 8–24, who each have one session, to avoid the much larger download for the longitudinal participants. This selection reflects protocol and compute constraints rather than subject-level performance. It limits inference to this subset. The notebook lists every subject ID explicitly.

TABLE I
Audited analysis contract

Property	Value
Subjects	8–24 (17)
Sessions / runs	One / two per subject
EEG channels	16 acquired; five decoder sites
Original sampling	512 Hz
Analysis sampling	128 Hz
MNE amplitudes	Volts; plots labelled in microvolts
Reference / ground	Left earlobe / Fz
Target / non-target code	33285 / 33286
Flashes per original run	80 target, 400 non-target
Total original flashes	16320

The audit confirms the expected sampling rate, channel count, conditions and class counts for every included run. Run labels come from each subject's metadata, rather than assuming a file number always identifies the same adaptive condition. Only the training phases enter this analysis. The labels Adaptive and Non-Adaptive describe the acquisition protocol; our offline LOSO model is not the original adaptive system.

A. Acquisition and Download Provenance

MOABB can prefer a mirror that provides only one training file. We therefore use the original Zenodo archives and retrieve their metadata and training MAT members with HTTP byte ranges. ZIP CRC checks verify archive-member extraction, and local SHA-256 manifests record the bytes. MOABB's original BI2013a reader still creates the MNE Raw objects. The downloader changes file retrieval, not the EEG parser or event definitions. Previously cached original files are reused. A server or network failure remains a reproducibility constraint.

B. Filtering and Sampling

Every run is filtered separately with a zero-phase Hamming-window FIR band-pass from 0.5 to 20 Hz. The high-pass removes slow drift and the low-pass suppresses out-of-band activity, including 50 Hz mains noise. A separate notch is omitted because the PSD and residual line-noise measurements assess attenuation after the low-pass. The notebook compares alternative high-pass cut-offs and a 30 Hz low-pass on the same example flashes, so changes in retained trials do not masquerade as a filter effect.

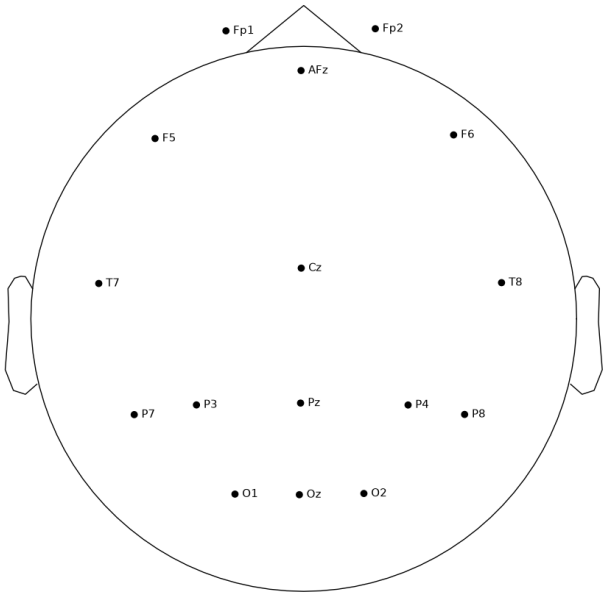


Fig. 1. Sensor layout from the MOABB/MNE montage. The decoder uses the fixed Cz/P3/Pz/P4/Oz subset. The remaining channels support inspection and artifact cleaning.

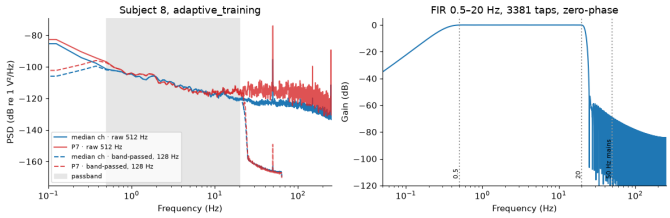


Fig. 2. Example subject: Welch spectral inspection before and after preprocessing. The spectrum and the filter response motivate drift removal and a 20 Hz low-pass. This is evidence from the notebook, not a simulated signal.

Resampling reduces 512 Hz to 128 Hz and resamples events together with the data. The notebook reports timing differences and signal error. A recording-edge guard uses the FIR half-support and the epoch bounds. It prevents the retained epochs from using unsupported edge samples. No filter support crosses a run boundary. Zero-phase filtering uses future samples, so the resulting system is explicitly offline.

III. Artifact Handling and Neurophysiology

Fp1 and Fp2 serve as ocular proxies because measured EOG is unavailable. These scalp sites also contain neural activity, so proxy correlation cannot establish an exclusively ocular source. The decoder excludes both sites.

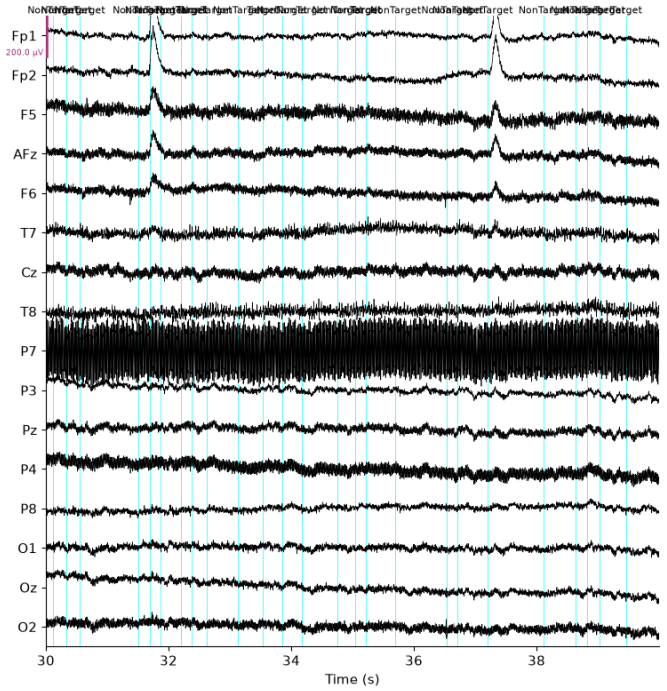


Fig. 3. Unfiltered example EEG. The viewer removes display offsets. Large frontal transients motivate ocular-artifact handling; numerical data retain their original units.

A. Cleaning and Quality Control

FastICA fits each unlabelled run on a 1–20 Hz copy at 128 Hz. The fitted unmixing operator is then applied to the 0.5–20 Hz signal. Up to 15 components are retained, with a reduction if flat channels reduce the available dimension. A component is excluded only when the proxy-correlation outlier rule and a frontal-map-power rule both flag it. Component evidence and before/after contrasts document the decision.

A channel is flat below 0.5 microvolt SD. A non-proxy channel is noisy when its cleaned SD exceeds three times, or falls below one third of, the non-proxy median. Channels remain visible in the audit. Runs with more than two bad channels and any flagged core decoder site stop analysis for review. There is no interpolation and no pooled channel-based feature selection.

Epochs extend from -100 to 800 ms. A 100 microvolt peak-to-peak rule rejects gross residual artifacts after the edge guard. We report class-specific counts rather than assuming rejection has no relation to target status. Across the full cohort, the edge guard removes 3 targets and 12 non-targets. Amplitude rejection then removes 80 targets and 329 non-targets. The analysis retains 2637 targets and 13259 non-targets, or 97.40 percent of all original flashes.

The baseline audit shows unequal preceding-flash class composition. Because responses overlap, baseline subtraction could introduce a class-dependent offset. We use no baseline subtraction in any condition.

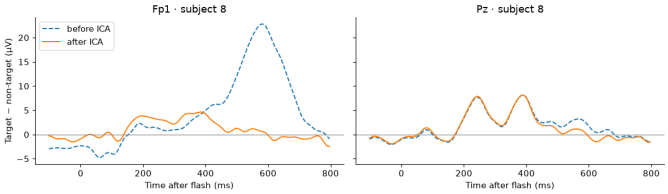


Fig. 4. Example artifact contrast before and after ICA. Proxy channels support cleaning but do not enter the classifier. Residual artifacts remain a limitation.

B. Expected P300 Response

Before decoding, target and non-target evoked responses are compared at Cz, Pz, P3, P4 and Oz. The fixed Pz window is 250–600 ms. The full-cohort mean target-minus-non-target window amplitude is 2.74 microvolts, with 16/17 subjects positive. Peak amplitudes, latencies, single-trial images and scalp distributions supplement the mean, so one number does not stand in for signal inspection.

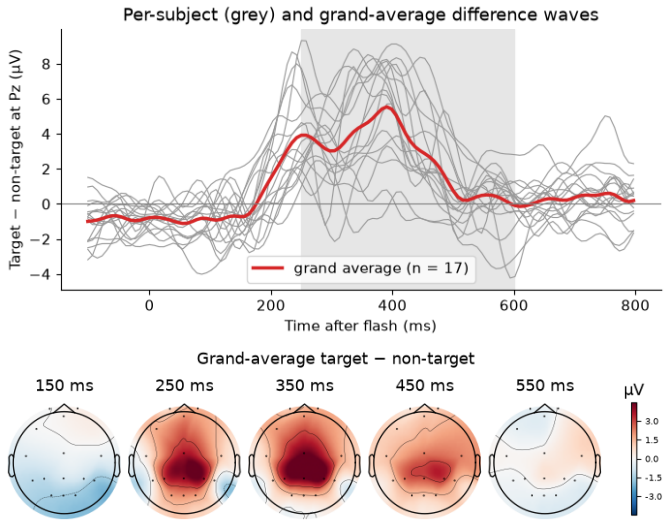


Fig. 5. Group target-minus-non-target response and scalp distribution from the executed notebook. Each subject contributes equally to the grand average. The fixed Pz window is 250–600 ms.

The response is consistent with P300 physiology. Cleaning does not establish neural specificity: residual ocular activity and neighbouring responses may contribute.

IV. Features, Models and Honest Evaluation

The feature map computes the mean amplitude in each of 16 consecutive 50 ms windows between 0 and 800 ms at the fixed five sites. Each epoch becomes an 80-dimensional vector. The window operation uses no estimated parameters or labels. It retains ERP timing information without using CSP, xDAWN, wavelets, neural networks or other methods outside the course's core scope.

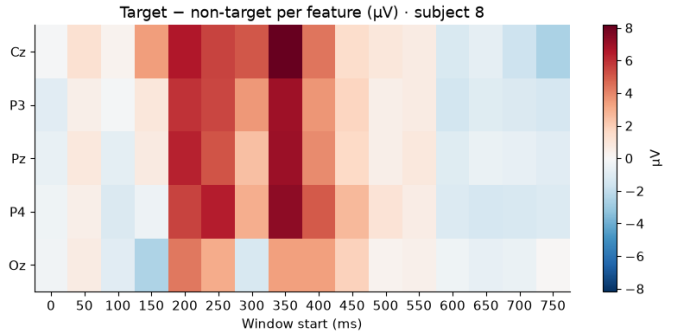


Fig. 6. Example subject: target-minus-non-target window means on the fixed five-channel montage. This descriptive heat map does not determine feature selection.

A. Training Boundary

One scikit-learn Pipeline combines window means, StandardScaler and LDA with an lsqr solver and analytic shrinkage. Scaling and covariance estimation fit only the training set. No hyperparameter search uses test scores. In the transfer loop, precomputing window means is equivalent to the stateless first Pipeline step; the scaler and LDA still refit for each training set. An assertion checks that the Within scores agree exactly with the separate grouped-CV implementation.

TABLE II
Controlled comparison for test run r of subject s

Condition	Training observations
Within	Other run of s
LOSO	Both runs of every subject other than s
LOSO-matched	External epochs with Within's target and non-target counts; ten seeded draws

Within evaluation uses LeaveOneGroupOut with run as the group. The transfer evaluation leaves an entire subject out. No random epoch split is used, and no epoch from a test subject enters the LOSO or matched training pool. Run-specific preprocessing keeps samples from crossing the split, although unlabelled cleaning sees the complete test run. The fixed montage ensures that test-subject channel flags cannot alter training.

B. Metrics and Chance

ROC AUC is primary because the original classes have a one-to-five ratio. AUC measures target/non-target ranking and has a chance reference of 0.5. Balanced accuracy is also reported for Within, using LDA's default decision rule without threshold tuning. Mean and sample SD are reported across the two Within folds and across subject means. These two kinds of spread answer different questions and must not be interchanged.

Within chance estimates use 200 label permutations restricted within run. LOSO chance estimates shuffle held-out labels against a fixed score vector 1,000 times, avoiding unnecessary refits. These shuffles retain class counts but do not preserve every dependency in the flash sequence. Because epochs overlap, the resulting per-run p-values are descriptive diagnostics rather than exact randomization inference. Above-chance group evidence is assessed separately at subject level.

C. Paired Statistical Comparison

Each subject contributes its mean over the two held-out runs. The runs do not count as independent people. H1 compares LOSO subject means with 0.5. H2 and H3 compare paired subject means in the predicted direction. We report Wilcoxon p-values, median paired differences and matched-pairs rank-biserial effects, excluding zero differences from effect-size ranks. Holm correction covers the three primary hypotheses. The LOSO-versus-matched comparison and ERP correlations remain exploratory.

A sensitivity analysis evaluates subjects 11–24 separately. Pilot subjects can still appear in training pools for these new users, which is permitted for a generic decoder. They do not contribute test outcomes to the sensitivity summary. This guards against treating the development participants as wholly unseen evaluation cases.

V. Full-Cohort Results

Table III gives the exact pairing underlying the hypothesis tests. Across subject means, Within AUC is 0.875 ± 0.056, LOSO-matched is 0.724 ± 0.071, and LOSO is 0.810 ± 0.072. These scores come from the executed notebook and exported CSVs. The median Within-minus-LOSO gap is 0.072, compared with a median Within fold SD of 0.018. The latter uses just two folds per subject and is an imprecise measure of run variability.

TABLE III
Per-subject mean ROC AUC across two test runs

Subject	Within	Matched	LOSO
8	0.884	0.765	0.812
9	0.883	0.821	0.906
10	0.891	0.662	0.770
11	0.913	0.721	0.777
12	0.869	0.784	0.856
13	0.861	0.652	0.738
14	0.832	0.626	0.752
15	0.929	0.700	0.862
16	0.932	0.835	0.916
17	0.812	0.609	0.671
18	0.961	0.837	0.926
19	0.805	0.663	0.714
20	0.759	0.671	0.773
21	0.960	0.758	0.848
22	0.884	0.742	0.849
23	0.882	0.737	0.789
24	0.815	0.729	0.809
Mean	0.875	0.724	0.810
SD	0.056	0.071	0.072

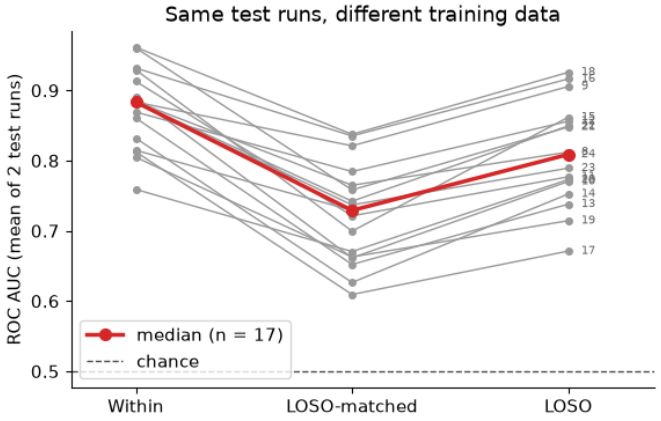


Fig. 7. Each line connects one subject across conditions. Test epochs are identical within a subject/run comparison. LOSO-matched averages ten class-count-matched draws.

Under the stated analysis, H1 is supported, H2 is supported, and H3 is supported. The conclusions use adjusted p-values rather than selecting whichever unadjusted comparison appears strongest. A supported directional difference would establish a cohort-level ordering, not equivalence for every individual or a guaranteed benefit for a future user. A non-significant comparison would indicate insufficient evidence under this analysis, not proof of equal performance.

TABLE IV
Primary tests: $n = 17$ subject means

Test	Median gap	r_{rb}	Raw p	Holm p
H1	0.309	1.000	7.6e-06	2.3e-05
H2	0.072	0.895	0.00019	0.00019
H3	0.142	1.000	7.6e-06	2.3e-05

TABLE V
Sensitivity excluding development subjects from evaluation

Test	n	Median gap	Holm p
H1	14	0.299	0.00018
H2	14	0.073	0.00031
H3	14	0.144	0.00018

The matched control addresses one practical alternative explanation. It preserves Within's class-specific training counts, so an observed gap cannot be attributed only to the total number of labelled epochs. The external examples come from several participants, however, and thus also differ in diversity and dependence structure. Ten draws quantify sampling variation, but do not remove those design differences.

VI. Interpretation, Limits and Reproducibility

A. What the Result Supports

The main contribution is a controlled comparison with a common in-scope decoder, identical test observations and explicit training boundaries. The full study extends the pilot to the declared cohort and reports both individual outcomes and paired comparisons. It asks how far a fixed linear ERP representation transfers, rather than attributing a score change to simultaneous changes in classifier, preprocessing and training quantity. The direction and magnitude of each result are reported even when they do not meet a significance threshold.

The ERP-property plots explore why some users may be easier to decode. Amplitude, latency and similarity to other users' grand average are measured descriptively. Similarity uses only the other subjects to form its template. Nevertheless, these associations use labelled test responses retrospectively, involve several comparisons, and cannot guide a prospective decoder without a separate validation procedure.

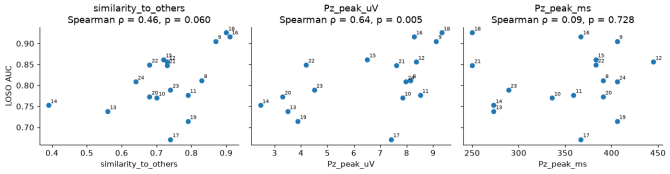


Fig. 8. Exploratory relationships between LOSO AUC and ERP properties. These associations are descriptive and do not feed the decoder or establish causal mechanisms.

B. Limits

The cohort includes 17 single-session users from one acquisition protocol. It does not establish transfer to different equipment, clinical populations, days or tasks. All three conditions are evaluated offline. Zero-phase filters and per-run ICA require future unlabelled samples, and therefore cannot establish immediate plug-and-play interaction. We decode single flashes rather than complete game commands. No character accuracy, information-transfer rate or online device-control result follows from AUC.

Overlapping flashes weaken the exchangeability assumption behind unrestricted label permutation. Subject-level comparisons reduce pseudoreplication, but the LOSO models share training subjects, so fitted-model errors are dependent. We present Wilcoxon statistics as conventional summaries of this fixed cohort, not exact guarantees for a new population. Two runs also provide limited information about day-to-day stability. Residual artifacts, pilot-informed decisions and the single physiological montage remain further limits. A future preregistered study should lock choices on a separate development cohort and test an independent acquisition.

C. Execution and Versions

Seed 613 and IDs 8–24 are explicit. The restarted Google Colab CPU session used Python 3.13.16. It completed in 7.41 minutes including setup checks and all data downloads; analysis after imports took 7.27 minutes. The dataset cache was initially empty. Pinned packages had been installed before a required NumPy restart. Request pacing and Retry-After handling resolved initial Zenodo rate limits. All notebook assertions passed. Saved notebook: https://colab.research.google.com/drive/19qYoD5urhaAcdMhhokzYAZPZuN2ra2Rh. CSVs and the manifest preserve exact results and versions.

TABLE VI
Exact direct dependency versions

Package	Version
mne	1.13.2
moabb	1.7.2
numpy	2.5.4
scipy	1.18.1
pandas	2.3.3
scikit-learn	1.9.1
matplotlib	3.11.2
PyYAML	6.0.3
remotezip	0.12.6
threadpoolctl	3.7.0

D. Authorship

The topic was registered on 15 October 2026. Lead and review roles and dated contributions are recorded in the contribution log and the Git history submitted with this report.

E. Conclusion

With fixed ERP features and a linear decoder, the full-cohort mean AUCs are 0.875 ± 0.056 for Within, 0.724 ± 0.071 for matched transfer and 0.810 ± 0.072 for LOSO. The primary comparisons are H1: Holm p = 2.3e-05; H2: Holm p = 0.00019; H3: Holm p = 2.3e-05. These results characterize offline transfer in this cohort. They support discussion of calibration burden while preserving the limits of a retrospective dataset analysis.

Appendix A
AI Tools Disclosure

OpenAI Codex was used for code review, debugging, API lookup, implementation changes to the notebook helpers, statistical checks, plotting, and drafting text for the report and slides. Claude Code (Anthropic) was used to check the submission against the assignment requirements, to edit report and README wording, and to write the slide generator. No AI tool generated results, citations or dataset facts: every number in this report is exported from the executed notebook, and each citation was checked against its source. All authors reviewed the submitted code and text and can explain it.

References

[1] E. Vaineau, A. Barachant, A. Andreev, P. Rodrigues, G. Cattan, and M. Congedo, "Brain Invaders Adaptive versus Non-Adaptive P300 Brain-Computer Interface dataset," arXiv:1904.09111, 2019. https://arxiv.org/abs/1904.09111.

[2] A. Barachant and M. Congedo, "A Plug&Play P300 BCI Using Information Geometry," arXiv:1409.0107, 2014. https://arxiv.org/abs/1409.0107.

[3] J. Kim and S.-P. Kim, "A Plug-and-Play P300-Based BCI With Zero-Training Application," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 33, pp. 3443–3454, 2025, doi: 10.1109/TNSRE.2025.3603979.

[4] B. Abibullaev, "ROBT613 BCI Workshop: Brain–Computer Interfaces with MNE-Python," notebooks 00–06, accessed Oct. 2026. https://berdakh.github.io/ROBT613-BCI-Workshop/, https://github.com/berdakh/ROBT613-BCI-Workshop.

[5] V. Jayaram and A. Barachant, "MOABB: trustworthy algorithm benchmarking for BCIs," J. Neural Eng., vol. 15, no. 6, 066011, 2018.